

Digital Twins for Protecting Cultural Heritage Against Climate Change

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Cyber–physical systems help toward mitigating the impact of climate change on cultural heritage sites and monuments by utilizing cultural heritage digital twins, coupled with artificial intelligence/machine learning algorithms, facilitating degradation detection and triggering appropriate actions.

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Cultural heritage (CH) is considered one of the most valuable assets around the world, providing links to our past, contributing to shaping our vision for the future, as well as being a means of promoting our values and way of life. However, CH faces increased risks and uncertainty as regards its protection and maintenance. At the same time, climate change poses threats to CH, in addition to anthropogenic factors and natural disasters, mandating us to devise innovative and sustainable ways to protect and preserve it. Aside from the dangers related to climate change that threaten CH, a recent report[1] by the International Council on Monuments and Sites stresses the role of CH, and the cultural and social values associated with it, in the transition to a more sustainable future and adaptation

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to climate change, together with the need to accelerate the adoption of mitigation measures to protect it. In recent years, a wave of new technologies has surfaced to help digitize material assets and bridge the real and the digital world, which were adapted and applied also to the field of CH. In this context, several methods[2] are used with respect to 3D digitization of tangible heritage, which is currently the most common form of such processes.

Given the growing challenges for CH, new technologies and practices need to be applied to improve on the current ones, as well as offer new possibilities on aspects that have not been covered adequately thus far; such aspects include monitoring the structural health of remote monuments and quantifying the effect of climate change on CH in coastal areas or underwater sites. There is also the potential of artificial intelligence (AI) to revolutionize the ways in which we digitize, monitor, and protect CH. AI can be used to assess the state of tangible CH or simulate the effect of climate change. Sensor-equipped drones can be used to produce 3D representations of monuments and sites at a speed that was unimaginable just a few years ago, while new types of sensors and nondestructive monitoring techniques can offer higher precision and reliability than existing approaches.

DIGITAL TWINS IN CH

CH digital twins (CHDT) have begun to appear for CH protection and maintenance, incorporating principles from the use of DTs in other domains, such as smart manufacturing. DTs have progressed from their first use by the National Aeronautics and Space Administration for simulating space modules in orbit, to holistic representations of more complex processes and systems in recent years. Historically, the term *digital twin* was first mentioned in 2003 in the context of product lifecycle management. DTs were quickly adopted in smart manufacturing and Industry 4.0, as well as in other domains like smart cities. DTs are continuously evolving, and the term is used in more flexible ways.

With respect to CHDTs, there seems to be a lack of a concise definition of what they are and what they are used for. Historically, in the case of built CH (BCH), the CH community has been using the building information modeling (BIM) concept for years, extending it to the historic BIM (HBIM). Although the CH community has started claiming to use DTs, in practice the HBIM approach continues to be the one used.[3]

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CHDTs could be seen as an evolution of HBIM, the latter providing the basic digitalization technology, combined with simulation capabilities, data fused from Internet of Things (IoT) deployments, and AI-based predictions, offering a computational model that "continuously learns from the physical asset and updates itself accordingly."[3] In other words, a CHDT focuses more on its actual physical counterpart than HBIM, by monitoring and simulating its behavior, integrating real-time data, and utilizing data analytics and AI for implementing services, such as predictive maintenance.

In addition, and with respect to climate change and its effects on CH, we consider several aspects as important elements of future CHDTs:

- *Climate change and environmental modeling:* Satellite-based monitoring will be key to assessing CH vulnerability against climate change-related factors; using such data, more reliable models can be produced, capturing the long- and short-term effects of climate change, waves, and floods on CH.
- *Structural health monitoring:* To provide comprehensive structural assessment, developments for global and local damage identification in CH sites should be explored, utilizing tools based on AI/machine learning (ML), IoT sensors, photo-acoustic methods, and data fusion to continuously monitor CH assets and update their digital counterparts.
- *Multimodal degradation identification:* New tools for automatic degradation detection should be used to facilitate tasks that traditionally require huge effort and expert knowledge, for example, by automating the process of monitoring the degradation status of a large-scale CH site.
- *Spectroscopic metrology for chemical analysis of CH artifacts:* Spectroscopic techniques can facilitate the analysis of complex organic and inorganic materials. They can also help track the status of aspects, such as pigments, providing valuable input for modern endangered CH assets.
- *Citizen science for CH monitoring:* Ambitious goals or policies

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cannot rely on technologies alone; citizens and social innovation could be a key factor toward turning political commitment into reality. Citizens and experts can provide input, for example, through smartphones to monitor degradation or annotate image datasets.

Figure 1 provides an illustration of an example scenario, where a variety of methods (including satellites, drones, IoT sensors, sensor-equipped buoys, and expert/visitor smartphones) are used to produce a digital "living" model of a CH site situated at a coastal area. The digitized site can also include underwater areas, while the modeling of sea waves and currents can be crucial as well. The data collected can be used in local climate and sea wave/currents models to produce macromodels for simulating the effect of climate change at large, while IoT data and 3D models work at a micro scale to continuously feed data to the CHDT and, together with the macromodels, produce more accurate predictions or suggestions to the experts.

THE WARMEST SOLUTION

The WARMEST solution[6] is an innovative example solution in CH protection and maintenance, addressing several aspects. It optimizes maintenance in cultural and natural heritage sites through the introduction of novel technologies and tools to collect and analyze data, as well as a decision support system. Part of the solution is a deep multimodal degradation identification component. As mentioned above, such a process can be used to automatically detect parts of structures or artifacts in CH sites that have some kind of defect/degradation, thus helping to automate the process of monitoring BCH at scale and accelerate mitigation actions to protect it. For instance, consider a scenario where a drone is used to periodically scan a large CH site that is threatened by climate change, and the data are then fed to a system to pinpoint areas of concern to experts.

More specifically, the system focuses on generating differential 3D saliency maps. Saliency maps correspond to textures that highlight regions of an object/site on which our eyes focus first and reflect a degree of importance to the human visual perception system, thus segmenting an image and making it easier to analyze. Differential saliency maps are calculated over 3D models that have been reconstructed via structure-from-motion techniques (SfM)[2] in two different time instances and capture differences on the saliency features that usually correspond to degradation. Those models have been originally deployed for generating 3D saliency maps using baseline traditional methods,[4] which are later used for training deep neural networks that will facilitate saliency extraction in a cost-efficient way. After the completion of the training process, the trained model can be used to extract saliency maps automatically



FIGURE 1. A high-level view of the parts used to create a DT for a coastal CH site.

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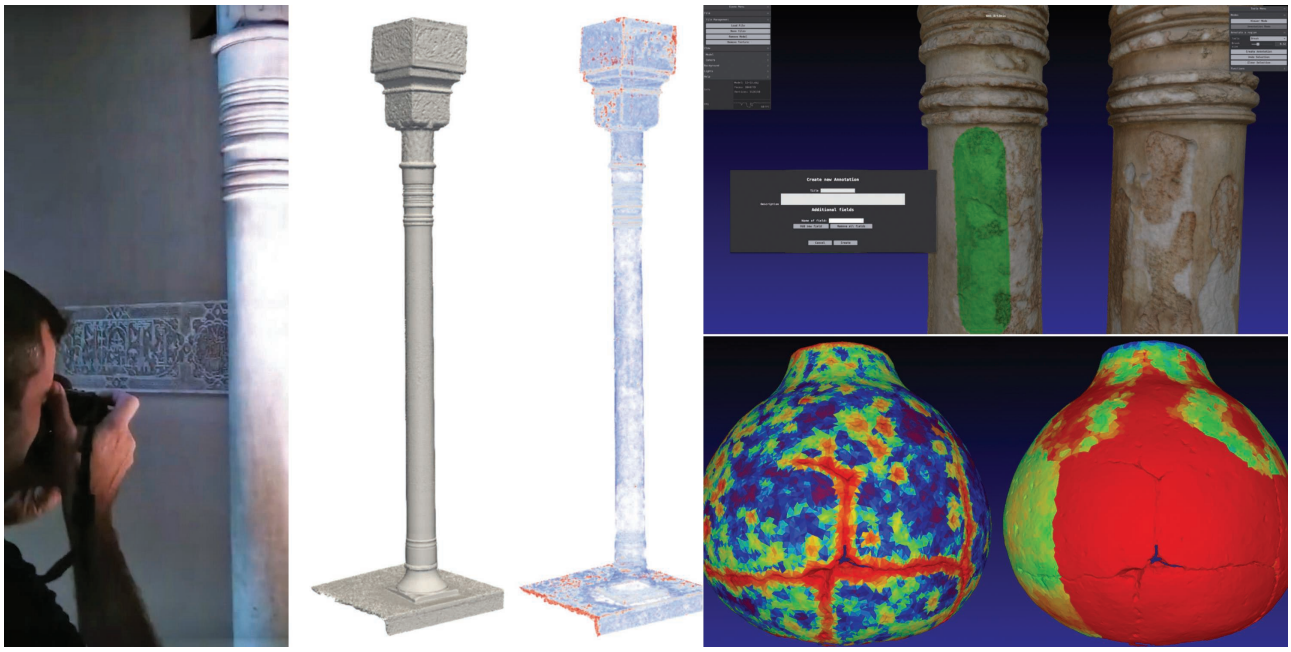


FIGURE 2. Examples showcasing the process pipeline utilized in WARMEST.

for any other new 3D model. To mitigate the complexity for both data generation and inference, the entire mesh is initially partitioned into a number of overlapping patches of equal size. Subsequently, two distinct steps are used to estimate the spectral and geometrical saliency features[5] and the combination of these two features yields the final saliency values.

Once the saliency mapping is obtained, it can be utilized as input to train/fine-tune novel deep neural networks, which subsequently help to identify regions of interest in the scanned cultural sites/artifacts. The solution is pragmatic and effective, as demonstrated via a series of experiments conducted using dense SfM 3D models reconstructed from captured imagery at two important European CH sites: the Patio de Leones of the Alhambra complex in Granada, Spain and the cloister of Santa Croce in Florence, Italy.

Figure 2 provides an illustration of the process: first, a set of photos of a column inside the Alhambra complex in Granada, Spain, are taken, which are then used to produce an accurate 3D model. This model is then fed to the system, which produces the saliency map and pinpoints areas of interest. A web-based tool is available to manually annotate the models or edit the system-produced annotations. The same method can be applied to artefacts of different sizes, for example, a ceramic pot. Such methods can simplify the process pipeline for CH experts (for example, archaeologists), as well as be used as input models to other tools, thus completing the CHDT picture.

Shielding CH from the effects of natural and anthropogenic hazards should be part of current and future actions. The utilization of CHDTs has enormous potential to transform the sector, facilitating and accelerating processes that were either performed manually in the past or were based on static models. The need to mitigate the effects of climate change and natural hazards will lead to solutions exploiting cyberphysical systems, AI/ML, and DTs that will become part of the toolbox of experts in this domain.

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